

Vertex crossings and boundary regimes in a symmetric Markov multinomial model

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Abstract

We study occupation counts of a stationary Markov chain that repeats its current state with probability p and otherwise chooses uniformly among $d - 1$ alternatives. For fixed d , we compare the probability of an individual count vector with that of a vertex, where every visit is to one state. A positive refresh expansion gives a relative Bessel approximation uniform over all positive compositions, including vectors approaching simplex boundaries. We obtain explicit crossing expansions for balanced faces and interior compositions, and a rare-coordinate crossover at $N/(\log N)^2$ when at least two coordinates are macroscopic. Near a vertex, a product of finite Laguerre polynomials determines the leading crossing uniformly for arbitrary rare counts with total $A = o(b)$, where b is the dominant count; the relative root error is $O_\ell(\sqrt{(A+1)/b})$. This root approximation can hold even when its corresponding probability approximation diverges. We also determine spatial superlevel volumes and construct weak occupation fields that make every support size optimal in succession. Exact laws and ordinary normal limits are included as background. The proposed contributions are the uniform moving-parameter estimates and their boundary and modal consequences; historical priority remains to be established.

1 Introduction

A symmetric persistent chain provides a simple model in which the most likely occupation vector changes from a vertex to the interior as the persistence decreases. The comparison involves the probability of one specific vector, rather than the aggregate probability of a face. It is especially sensitive to coordinates that are small relative to the total number of visits. This paper studies those crossing locations for a fixed number of states, including count vectors approaching a vertex at arbitrary rates.

The model is a multistate version of the Galton-board process in Kagey's Problem 131 [1]. The binary analysis and its free/periodic boundary comparison are developed in the unpublished companion [2]. Here we state the multistate model explicitly and prove all estimates needed for the multistate results; no theorem from the companion is required for a proof in this paper. In geometric terms, the occupation vectors index the bins of a simplex board with fixed global directions and uniform switching. Other rules at physical boundaries can define different processes.

Our main estimates address three regimes. On a fixed simplex face, the balanced crossing lies in the common window $Nu = \log N - \frac{1}{2} \log \log N + O(1)$, with explicit constants depending on its support size. A relative Bessel bound uniform over every positive composition resolves the transition when additional coordinates have size $N/(\log N)^2$. Finally, if one count is b and the remaining counts have total $A = o(b)$, a single polynomial equation gives the crossing with relative

error $O_\ell(\sqrt{(A+1)/b})$, uniformly over their individual sizes. The last assertion remains valid in a regime where the associated probability approximation is not uniform. The modal consequences include spatial superlevel volumes and an explicit weak field for which all support sizes occur successively.

Relation to established results. The underlying law belongs to Markov multinomial theory [3, 4]. The exact finite sum and the ordinary normal limit recorded below are background, not novelty claims. The original Wang–Yang article has not been obtained; its bibliographic information is reported through Wang and Tang, and comparison with its full text remains part of the priority review. General continuous-time occupation densities on visited simplex faces are also established [5]. The square-root composition quantity appearing below is classical: if Q is the complete-graph continuous-time generator with off-diagonal entries one, then its symmetric occupation rate is

$$I_Q(\alpha) = -\langle \sqrt{\alpha}, Q\sqrt{\alpha} \rangle = d - \left(\sum_i \sqrt{\alpha_i} \right)^2.$$

Thus the exponent $(\sum_i \sqrt{\alpha_i})^2 - 1$ is the difference between the vertex rate and the rate at α . We do not claim this rate function as a new result. Standard Bessel expansions are taken from [6]; all discrete comparison estimates and root inversions used here are proved explicitly. The proposed contribution is their uniform control in moving persistence regimes and the resulting boundary and modal statements. The literature search has not established historical priority, and this draft is held for further mathematical and bibliographic review.

Organization and scope. Section 2 gives the exact law, elementary estimates, and fixed-persistence covariance and normal limit. Section 3 proves the face crossings and balancing theorem. Sections 4 and 5 treat boundary coordinates and the uniform near-vertex law. Section 6 treats superlevel sets and weak fields. The results do not give every finite-size global mode or cover a growing number of states. A finite-size example shows why a vertex/full-center dichotomy cannot hold without qualification.

2 The model and exact formulas

Fix $d \geq 2$ and let $(X_j)_{j \geq 1}$ be a Markov chain on $\{1, \dots, d\}$. The initial state is uniform. At each subsequent step the chain repeats its state with probability $p \in [0, 1]$ and moves to each other state with probability $r = (1 - p)/(d - 1)$. Put $q = 1 - p$ and $\sigma = p - r = (dp - 1)/(d - 1)$. Let $K_N = (K_{N,1}, \dots, K_{N,d})$ be the occupation vector over N visits, and write $P_N(k) = \mathbb{P}(K_N = k)$. For a word w of length $N \geq 1$ with j changes, its probability is $d^{-1}p^{N-1-j}r^j$. The vertex probability is $V_N = p^{N-1}/d$. For $p > 0$ put $u = r/p$; for a positive vector $n = (n_1, \dots, n_k)$ with total N , write

$$S_n(u) = \frac{P_N(n_1, \dots, n_k, 0, \dots, 0)}{V_N} = \sum_j W_n(j) u^j,$$

where $W_n(j)$ counts words on the specified k states with these occupations and j changes. The ratio is independent of the ambient number d once the positive coordinates are fixed. Set $P_0(0, \dots, 0) = 1$ separately.

Proposition 2.1 (d directions). *Let $P_N(k_1, \dots, k_d)$ be the probability that direction i is taken k_i times. Put $\sigma = p - \frac{q}{d-1} = \frac{dp-1}{d-1}$ and $\mathcal{R} = \sum_{i=1}^d \frac{x_i t}{1-\sigma x_i t}$. Then*

$$\sum_{N \geq 0} \sum_{k_1 + \dots + k_d = N} P_N(k_1, \dots, k_d) x_1^{k_1} \dots x_d^{k_d} t^N = 1 + \frac{\mathcal{R}/d}{1 - \frac{q}{d-1} \mathcal{R}}.$$

Proof. Let F_i be the generating function of words ending in state i , and $S = \sum_i F_i$. Conditioning on the last bounce, $F_i = x_i t (\frac{1}{d} + p F_i + \frac{q}{d-1} (S - F_i))$, that is $F_i(1 - \sigma x_i t) = x_i t (\frac{1}{d} + \frac{q}{d-1} S)$. Summing over i gives $S = \mathcal{R}(\frac{1}{d} + \frac{q}{d-1} S)$, and the generating function is $1 + S$. $\square$

For $d = 2$, setting $x_1 = x$ and $x_2 = 1$ gives the binary generating function studied in the companion [2].

Proposition 2.2 (finite bin probabilities in d directions). *Put $r = q/(d-1)$ and $\sigma = p - r$. For $N \geq 1$, let $k = (k_1, \dots, k_d)$ have nonnegative integer coordinates summing to N , and put $H = \{i : k_i > 0\}$. Sum over $m_i = 0$ for $i \notin H$ and $1 \leq m_i \leq k_i$ for $i \in H$, and set $M = \sum_i m_i$. Then*

$$P_N(k) = \frac{1}{d} \sum_m \frac{M!}{\prod_i m_i!} r^{M-1} \sigma^{N-M} \prod_{i \in H} \binom{k_i - 1}{m_i - 1}.$$

All exponents are nonnegative, and $0^0 = 1$. Set $P_0(0, \dots, 0) = 1$ separately.

Proof. With $z_i = x_i t$, Proposition 2.1 becomes the formal power series

$$1 + \frac{1}{d} \sum_{M \geq 1} r^{M-1} \left(\sum_i \frac{z_i}{1 - \sigma z_i} \right)^M.$$

Expand by the multinomial theorem and use, for $k \geq m \geq 1$,

$$[z^k] \left(\frac{z}{1 - \sigma z} \right)^m = \binom{k-1}{m-1} \sigma^{k-m}.$$

For $m = 0$ only $k = 0$ contributes. Multiplying the coefficients proves the formula. $\square$

At $d = 3$ this is a triple sum, with the support convention covering edge bins. At $p = 1/d$, $\sigma = 0$ and only $M = N$ survives, giving the ordinary multinomial law. At $p = 1$, only $M = 1$ survives, giving mass $1/d$ at each $N e_i$. The formula is valid also at $p = 0$; for $\sigma < 0$ its terms can cancel, so dynamic programming or exact arithmetic is preferable numerically. This geometry assumes fixed global directions and uniform switching to the other directions; a different physical boundary or orientation rule need not have this law.

2.1 Elementary bounds used below

For $x \geq 0$, the modified Bessel function used here is

$$I_1(x) = \sum_{j \geq 0} \frac{(x/2)^{2j+1}}{j!(j+1)!}.$$

Its coefficients give $I_1(x) \leq \sinh x \leq e^x$. We also use the standard expansion [6, Eq. 10.40.1]

$$I_1(x) = \frac{e^x}{\sqrt{2\pi x}} \left(1 - \frac{3}{8x} + O(x^{-2}) \right), \quad x \rightarrow \infty.$$

The Laguerre–Bessel comparison needed at boundaries is proved in Lemma 4.1; it is not imported from the binary companion.

Lemma 2.3 (clipped product deficit). *For any finite list of nonnegative real numbers $t_1, \dots, t_m$, with $(x)_+ = \max(x, 0)$,*

$$0 \leq 1 - \prod_{j=1}^m (1 - t_j)_+ \leq \sum_{j=1}^m t_j.$$

Proof. If some $t_j \geq 1$, the product is zero and the sum is at least one. Otherwise all factors lie in $[0, 1]$ and induction follows from $1 - (1 - a)(1 - b) = a + b - ab \leq a + b$. The empty product gives equality. $\square$

2.2 Covariance and the fixed-persistence normal limit

The following formulas are recorded for comparison with the moving regimes studied later. Their fixed- p normal limit is not a local approximation for individual occupation probabilities when p depends on N .

Remark 2.4 (covariance and the tetrahedral case). For the board with d directions of Proposition 2.1 let K_a be the number of bounces in direction a . The direction chain has transition matrix $\sigma I + (1 - \sigma)\frac{1}{d}J$ with J the all-ones matrix, so the indicator of direction a at time i and of direction b at time $i + m$ have covariance $\sigma^m \left(\frac{\delta_{ab}}{d} - \frac{1}{d^2} \right)$. Summing over the two time indices and evaluating the finite geometric sum gives

$$\text{Cov}(K_a, K_b) = \left(\frac{\delta_{ab}}{d} - \frac{1}{d^2} \right) \left(N \frac{1 + \sigma}{1 - \sigma} - \frac{2\sigma(1 - \sigma^N)}{(1 - \sigma)^2} \right), \quad \frac{1 + \sigma}{1 - \sigma} = \frac{d - 2 + dp}{d(1 - p)}.$$

These formulas hold for $p < 1$; at $p = 1$ the covariance is $N^2(\delta_{ab}/d - 1/d^2)$. The asymptotic multiplier of the multinomial covariance is $(d - 2 + dp)/(d(1 - p))$, which is $(1 + 3p)/(3(1 - p))$ for the tetrahedron.

Theorem 2.5 (simplex normal limit). *Let $d \geq 3$ and $0 \leq p < 1$, or $d = 2$ and $0 < p < 1$. Set*

$$\pi = (1/d, \dots, 1/d), \quad C = \frac{1}{d}I - \frac{1}{d^2}J, \quad c = \frac{1 + \sigma}{1 - \sigma} = \frac{d - 2 + dp}{d(1 - p)}.$$

Then $(K_N - N\pi)/\sqrt{N}$ converges in distribution to $\mathcal{N}_d(0, cC)$. This Gaussian is supported on the sum-zero hyperplane and is nondegenerate there.

Proof. Let $\Pi = \mathbf{1}\pi$ and $T = \sigma I + (1 - \sigma)\Pi$. Fix a real vector a and put $f_i = a_i - d^{-1} \sum_j a_j$, so $\pi f = 0$ and $Tf = \sigma f$. For $D(t) = \text{diag}(e^{itf_i})$ and $B(t) = TD(t)$, stationarity gives

$$\mathbb{E} \exp \left(it \sum_{j=1}^N f(X_j) \right) = \pi D(t) B(t)^{N-1} \mathbf{1}.$$

At zero, $B(0) = T$ has the simple eigenvalue 1 and all other eigenvalues equal σ , with $|\sigma| < 1$. Analytic perturbation gives an eigenvalue $\lambda(t)$ and right eigenvector $h(t)$ with $\lambda(0) = 1$, $h(0) = \mathbf{1}$ and $\pi h(t) = 1$. The complementary spectral part is uniformly exponentially small for sufficiently small t : choose a circle of radius $\eta < 1$ enclosing the complementary spectrum but not $\lambda(t)$ and use the resolvent formula for matrix powers. Thus the characteristic function equals $b(t)\lambda(t)^{N-1} + O(\eta^N)$ with $b(t) \rightarrow 1$.

Differentiate $Bh = \lambda h$ at zero. Since $B'(0) = iT \operatorname{diag}(f)$,

$$\lambda'(0) = 0, \quad (I - T)h'(0) = i\sigma f, \quad h'(0) = \frac{i\sigma}{1 - \sigma}f.$$

Differentiating again and multiplying by π gives

$$\lambda''(0) = \pi B''(0)\mathbf{1} + 2\pi B'(0)h'(0) = -\frac{1 + \sigma}{1 - \sigma}\pi(f^2) = -c\pi(f^2).$$

Hence $\lambda(t) = 1 - c\pi(f^2)t^2/2 + O(t^3)$. Substitute $t = s/\sqrt{N}$ to obtain the scalar normal characteristic function with variance $c\pi(f^2) = ca^\top Ca$. Lévy's theorem and the Cramér–Wold device give the result. $\square$

For $d = 3$ the first two centered coordinates have limiting covariance

$$\frac{1 + 3p}{27(1 - p)} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix};$$

the third is minus their sum. Unlike the binary case, $p = 0$ is included for $d \geq 3$. At $d = 2, p = 0$ bounded alternation gives the zero limit after division by $\sqrt{N}$. At $p = 1$ the centered sequence on this scale is not tight. This theorem asserts weak convergence, not a local approximation for individual bin probabilities.

3 Face-center thresholds on a simplex

We retain the symmetric d -direction model of Proposition 2.1: the initial direction is uniform, a repeat has probability p , and each specified change has probability $r = (1 - p)/(d - 1)$. This specification matters; the results below do not cover every physical interpretation of a tetrahedral board. For $2 \leq k \leq d$ and $N \geq k$, let $C_{N,k}(p)$ denote the probability of one *balanced face bin*: its k positive coordinates differ by at most one, and its other coordinates vanish. All such bins have the same probability. Let $V_N(p) = p^{N-1}/d$ be the probability of one vertex bin, and put

$$u = \frac{1 - p}{(d - 1)p}, \quad a_k = \frac{1}{2} \log(4\pi) - \frac{k + 1/2}{k - 1} \log k.$$

The ratio $C_{N,k}/V_N$, viewed as a function of u , depends on k and N but not on the ambient number d of directions.

Theorem 3.1 (face-center crossings). *For every $2 \leq k \leq d$ and $N \geq k$, there is exactly one $p = p_{N,k}^{(d)} \in (0, 1)$ for which $C_{N,k}(p) = V_N(p)$. The face center is more likely below this persistence and less likely above it. The sequence $p_{N,k}^{(d)}$ is strictly increasing in N , and*

$$p_{k,k}^{(d)} = \frac{1}{1 + (d - 1)(k!)^{-1/(k-1)}}.$$

Writing $L = \log N$ and $u_{N,k} = (1 - p_{N,k}^{(d)})/((d - 1)p_{N,k}^{(d)})$, for fixed k we have

$$Nu_{N,k} = L - \frac{1}{2} \log L + a_k + \frac{\frac{1}{4} \log L - \frac{1}{2} a_k + \frac{1}{8}}{L} + O_k\left(\frac{(\log L)^2}{L^2}\right). \quad (1)$$

Consequently, $N(1 - p_{N,k}^{(d)})$ equals $(d - 1)$ times the right-hand side of (1), with the same error order for fixed d .

In particular, for the full center $k = d$,

$$N(1 - p_{N,d}^{(d)}) = (d-1) \log N - \frac{d-1}{2} \log \log N + c_d + o(1), \quad c_d = \frac{d-1}{2} \log(4\pi) - \frac{2d+1}{2} \log d.$$

For $d = 2$, $a_2 = \frac{1}{2} \log(\pi/8)$, and all the displayed coefficients agree with the binary formula in the unpublished companion [2]. The proof below is independent of that formula and treats every fixed face dimension in the same moving regime.

Theorem 3.2 (common critical window). *Fix d and put*

$$u_N(t) = \frac{\log N - \frac{1}{2} \log \log N + t}{N}, \quad p_N(t) = \frac{1}{1 + (d-1)u_N(t)}.$$

Uniformly for t in any fixed compact interval, for each $2 \leq k \leq d$,

$$\frac{C_{N,k}(p_N(t))}{V_N(p_N(t))} \longrightarrow \exp((k-1)(t - a_k)). \quad (2)$$

Moreover $a_2 > a_3 > \dots$. Thus the face centers cross the vertex height at distinct constant offsets within a common window of persistence width $O(1/N)$.

We first prove the exact crossing assertions and then establish a uniform local estimate that proves both asymptotic theorems. No fixed- p central limit theorem is used to infer probabilities in this moving regime.

Proof of the exact assertions in Theorem 3.1. For a positive count vector $n = (n_1, \dots, n_k)$, let $W_n(j)$ be the number of words with those counts and exactly j changes. Its ratio to a vertex probability is

$$S_n(u) = \sum_j W_n(j) u^j.$$

To count $W_n(j)$, choose a proper word of $j+1$ run colors, meaning that successive colors differ. If color i appears m_i times, its run lengths can be chosen in $\binom{n_i-1}{m_i-1}$ ways. Thus $W_n(j)$ is a sum of products of these binomial coefficients. Every coefficient is nonnegative; the first nonzero one is $W_n(k-1) = k!$. Hence S_n is strictly increasing from zero to infinity on $u \geq 0$ and crosses one exactly once. At $p = 0$, cycling through the k colors and truncating after N positions gives a word with balanced counts and no successive repeat, so its probability is positive while the vertex probability is zero. At $p = 1$ the reverse strict inequality holds.

Choose the balanced count vectors as N increases by incrementing one of their minimum coordinates. Each coefficient $W_n(j)$ is then nondecreasing. The coefficient of u^k is

$$W_n(k) = \frac{k!(k-1)}{2} \sum_i (n_i - 1) = \frac{k!(k-1)}{2} (N - k).$$

Indeed, exactly one run color appears twice; for each such color there are $(k+1)!/2 - k! = k!(k-1)/2$ proper arrangements of the multiset of run colors. This coefficient increases strictly with N , so the positive root decreases strictly with N . The substitution $p = 1/(1 + (d-1)u)$ proves the claimed monotonicity. If $N = k$, every coordinate equals one and $S_n(u) = k!u^{k-1}$, giving the initial value. Notice also that every root satisfies $u \leq (k!)^{-1/(k-1)} < 1$, so all crossings lie in the positive refresh regime used below. $\square$

Lemma 3.3 (positive integral and discrete comparison). *For fixed $k \geq 2$, define*

$$F_k(x) = \sum_{m_1, \dots, m_k \geq 1} M! \prod_{i=1}^k \frac{x^{m_i}}{m_i! (m_i - 1)!} = \int_0^\infty e^{-t} [\sqrt{xt} I_1(2\sqrt{xt})]^k dt, \quad M = \sum_i m_i. \quad (3)$$

Let n be a balanced positive count vector of total N , put $h = N/k$, and, for $0 < u < 1$, put $x = hu/(1-u)$. With $\mathcal{D} = x d/dx$,

$$0 \leq 1 - \frac{S_n(u)}{(1-u)^{N-1} h^{1-k} F_k(x)/x} \leq \frac{\mathcal{D}^2 F_k(x)}{h F_k(x)} = O_k\left(\frac{(1+x)^2}{h}\right). \quad (4)$$

In particular the relative error is $O_k((\log N)^2/N)$ when $x = O(\log N)$.

Proof. The equality in (3) follows from the Bessel power series and Tonelli's theorem, using $M! = \int_0^\infty e^{-t} t^M dt$. The integral is finite for every positive x : the bound $I_1(y) \leq e^y$ for $y \geq 0$ reduces its tail to a polynomial times $\exp(-t + 2k\sqrt{xt})$. Thus the positive series and all its termwise derivatives converge on compact subsets of $x > 0$.

Set $\sigma = p - r = p(1-u)$ and $v = r/\sigma = u/(1-u)$. Proposition 2.2, restricted to the positive coordinates, gives

$$S_n(u) = (1-u)^{N-1} \sum_{m_i \geq 1} M! v^{M-1} \prod_i \frac{\binom{n_i-1}{m_i-1}}{m_i!}. \quad (5)$$

Define

$$A_{i,m} = \frac{(m-1)!}{h^{m-1}} \binom{n_i-1}{m-1} = \prod_{j=1}^{m-1} \left(1 - \frac{h - n_i + j}{h}\right)_+.$$

Since $h - n_i > -1$, these are products of factors in $[0, 1]$, including beyond their finite support. The elementary product-deficit bound in Lemma 2.3 gives

$$0 \leq 1 - \prod_i A_{i,m_i} \leq \frac{1}{h} \sum_i \frac{m_i(m_i+1)}{2} \leq \frac{M^2}{h}.$$

Replacing the binomial factors in (5) by $h^{m_i-1}/(m_i-1)!$ gives the denominator in (4). Summing the deficit bound proves its first inequality and the displayed explicit upper bound.

To bound the quotient of derivatives, for $x \geq 1$ use $M^2 \leq 3x^2(1+1/x)^M$. Indeed, $\log(1+1/x) \geq 1/(2x)$ and $\sup_{y \geq 0} y^2 e^{-y/2} = 16/e^2 < 3$. Positivity of the coefficients gives

$$\frac{\mathcal{D}^2 F_k(x)}{F_k(x)} \leq 3x^2 \frac{F_k(x+1)}{F_k(x)} = O_k(x^2),$$

where the last step follows from Lemma 3.4 below. For $0 < x \leq 1$, the quotient extends continuously to $x = 0$ with value k^2 , since $F_k(x) = k!x^k + O_k(x^{k+1})$. This proves the asserted bound without differentiating an asymptotic expansion. $\square$

Lemma 3.4 (Laplace expansion). *For each fixed $k \geq 2$, as $x \rightarrow \infty$,*

$$F_k(x) = \frac{k^{k/2+1}}{(4\pi)^{(k-1)/2}} x^{(k+1)/2} e^{k^2 x} \left(1 - \frac{k-1}{8kx} + O_k(x^{-2})\right). \quad (6)$$

Proof. In (3) set $w = \sqrt{t}$ and $a = k\sqrt{x}$. The exponent becomes $-w^2 + 2k\sqrt{x}w = k^2x - (w - a)^2$. First restrict to $|w - a| \leq \sqrt{x}/2$. There $2\sqrt{x}w$ is bounded below by $(2k - 1)x$, so the standard Bessel expansion [6, Eq. 10.40.1] applies uniformly:

$$[\sqrt{x}wI_1(2\sqrt{x}w)]^k = \frac{x^{k/4}w^{k/2}e^{2k\sqrt{x}w}}{(4\pi)^{k/2}} \left(1 - \frac{3k}{16\sqrt{x}w} + O_k(x^{-2})\right).$$

On the complementary set, the elementary exponential bound for I_1 dominates the integrand by $2x^{k/2}w^{k+1}e^{k^2x-(w-a)^2}$. Gaussian tail estimates therefore show that its integral is at most e^{k^2x-cx} times a fixed power of x , for some $c > 0$. It is exponentially smaller than the main term.

Put $\alpha = k/2 + 1$. Apart from that negligible tail, the integral is

$$\frac{2x^{k/4}e^{k^2x}}{(4\pi)^{k/2}} \int_{|w-a| \leq \sqrt{x}/2} e^{-(w-a)^2} w^\alpha \left(1 - \frac{3k}{16\sqrt{x}w} + O_k(x^{-2})\right) dw.$$

Taylor-expand w^α about a . The odd terms integrate to zero on this symmetric interval, the quadratic term contributes $\alpha(\alpha - 1)/(4a^2)$ relatively, and the fourth-order remainder is $O_k(a^{-4})$: the interval stays a fixed positive fraction of a away from zero and all Gaussian moments are finite. Extending the centered Gaussian integrals to the real line changes them exponentially little. For the term involving $w^{\alpha-1}/\sqrt{x}$, keeping its constant term suffices, because its quadratic correction has relative order x^{-2} . Since $\int e^{-y^2} dy = \sqrt{\pi}$ and $\int y^2 e^{-y^2} dy = \sqrt{\pi}/2$, the total relative correction is

$$\frac{\alpha(\alpha - 1)}{4k^2x} - \frac{3}{16x} = -\frac{k - 1}{8kx}.$$

The leading constant simplifies to the one in (6), completing the proof. $\square$

Proof of the asymptotic assertions in Theorems 3.1 and 3.2. Set $s = k - 1$, $z = Nu$, and $D_k = k^{k+1/2}/(4\pi)^{(k-1)/2}$. The preceding lemmas imply, uniformly on $c \log N \leq z \leq C \log N$ for fixed $0 < c < C$,

$$S_n(z/N) = D_k \frac{z^{s/2} e^{sz}}{N^s} \left(1 - \frac{s}{8z} + O_{k,c,C}(z^{-2} + z^2/N)\right). \quad (7)$$

Indeed, $x = z/(k(1 - u))$, and

$$k^2x + (N - 1) \log(1 - u) - sz = O_k(z^2/N).$$

The prefactor changes by $1 + O_k(z/N)$ on replacing x by z/k ; rounding of the balanced coordinates is covered by (4).

At $z = (1 - \varepsilon) \log N$ the ratio in (7) tends to zero, whereas at $z = (1 + \varepsilon) \log N$ it tends to infinity. The unique root therefore has $z/\log N \rightarrow 1$. Taking logarithms at the root and dividing by s gives

$$z + \frac{1}{2} \log z = L + a_k + \frac{1}{8z} + O_k(z^{-2} + z^2/N).$$

For an explicit inversion, put

$$d_L = -\frac{1}{2} \log L + a_k, \quad e_L = \frac{1}{8} - \frac{1}{2} d_L, \quad z_* = L + d_L + e_L/L.$$

Taylor expansion gives $z_* + \frac{1}{2} \log z_* - 1/(8z_*) = L + a_k + O_k((\log L)^2/L^2)$. The derivative of $z + \frac{1}{2} \log z - 1/(8z)$ is at least one for $z > 0$. The preceding root equation, with $z \sim L$, therefore gives $z - z_* = O_k((\log L)^2/L^2)$, proving (1). The conversion to physical persistence uses

$N(1-p) = (d-1)z/(1+(d-1)z/N) = (d-1)z + O_d(z^2/N)$. Substituting $z = L - \frac{1}{2} \log L + t$ in (7) proves (2), uniformly on compact t intervals.

For the ordering of constants, extend $g(x) = (x+1/2) \log x/(x-1)$ to $x \geq 2$. The sign of $g'(x)$ is the sign of $H(x) = 2x^2 - x - 1 - 3x \log x$. Here $H(2) = 5 - 6 \log 2 > 0$, and $H'(x) = 4x - 4 - 3 \log x > 0$ for $x \geq 2$, since its derivative is $4 - 3/x > 0$ and $H'(2) > 0$. Thus g increases strictly, so a_k decreases strictly. $\square$

3.1 Thresholds throughout the interior of a face

The same argument gives a spatially varying threshold throughout a face, not just at its center. All constants in the following statement use the *actual* fractions $\alpha_i = n_i/N$. This is necessary if the count vector varies with N without a specified rate of convergence.

Theorem 3.5 (interior composition thresholds). *Fix $k \geq 2$ and $\varepsilon > 0$. Let $n_1 + \dots + n_k = N$, with $\alpha_i = n_i/N \geq \varepsilon$, and define*

$$A = \sum_{i=1}^k \sqrt{\alpha_i}, \quad g = A^2 - 1, \quad s = k - 1, \quad D(\alpha) = \frac{A^{k/2+1} (\prod_i \alpha_i)^{-3/4}}{(4\pi)^{s/2}}.$$

Uniformly for these count vectors and $c \log N \leq z \leq C \log N$, with fixed $0 < c < C$,

$$S_n(z/N) = D(\alpha) \frac{z^{s/2} e^{gz}}{N^s} (1 + O_{k,\varepsilon,C}(z^{-1} + z^2/N)). \quad (8)$$

Let $u_n > 0$ be the unique root of $S_n(u) = 1$. Writing $L = \log N$,

$$Nu_n = \frac{s}{g} L - \frac{s}{2g} \log L - \frac{\log D(\alpha) + (s/2) \log(s/g)}{g} + O_{k,\varepsilon} \left(\frac{\log L}{L} + \frac{L^2}{N} \right). \quad (9)$$

These estimates are uniform on every compact subset of the relative interior of the face. For a fixed target α and integer counts $n_i = N\alpha_i + O(1)$, the constants may equivalently be evaluated at that target, without changing the error order.

Proof. The proper-run polynomial used above has nonnegative coefficients and first coefficient $k!u^{k-1}$ for every positive count vector, which gives the unique positive root. To obtain the approximation, put $v = u/(1-u)$ and $y = Nv$, and introduce

$$G_\alpha(y) = \int_0^\infty e^{-t} \prod_i [\sqrt{\alpha_i y t} I_1(2\sqrt{\alpha_i y t})] dt.$$

Replacing each binomial coefficient in (5) by $n_i^{m_i-1}/(m_i-1)!$ gives

$$S_n(u) = \frac{(1-u)^{N-1}}{v \prod_i n_i} G_\alpha(y) (1 + O_{k,\varepsilon}((1+y)^2/N)). \quad (*)$$

Here is a uniform justification of the error. The corresponding falling-factorial deficit is bounded by $\sum_i m_i(m_i-1)/(2n_i) \leq M^2/(2\varepsilon N)$. The positive-series argument in Lemma 3.3 bounds the weighted second moment of M by a constant times $(1+y)^2$. Uniformity follows from the uniform saddle expansion below, which bounds $G_\alpha(y+1)/G_\alpha(y)$ on the compact set $\{\alpha_i \geq \varepsilon, \sum_i \alpha_i = 1\}$; for bounded y , continuity and the positive leading coefficient give the same conclusion.

Set $w = \sqrt{t}$. The exponential factor of the integrand is

$$\exp(-w^2 + 2A\sqrt{y}w) = \exp(A^2y - (w - A\sqrt{y})^2).$$

Every α_i stays bounded away from zero, so the Bessel expansions, Gaussian tail estimates, and Taylor remainders in Lemma 3.4 are uniform here. Keeping their leading terms yields

$$G_\alpha(y) = \frac{A^{k/2+1}(\prod_i \alpha_i)^{1/4}}{(4\pi)^{s/2}} y^{(k+1)/2} e^{A^2y} (1 + O_{k,\varepsilon}(y^{-1})).$$

Substitute this into (*) and use $v \prod_i n_i = yN^{k-1} \prod_i \alpha_i$. For $u = z/N$ in the stated range, $y = z + O(z^2/N)$ and $(N-1)\log(1-u) = -z + O(z^2/N)$. This proves (8) with the specified constant.

On the compact set of compositions, both g and $D(\alpha)$ have positive lower and finite upper bounds. Testing the ratio at suitable constant multiples of $\log N$ brackets the unique root uniformly. Taking logarithms there gives

$$gz + \frac{s}{2} \log z = sL - \log D(\alpha) + O_{k,\varepsilon}(z^{-1} + z^2/N).$$

First $z = (s/g)L + O_{k,\varepsilon}(\log L)$; one substitution then gives (9). The derivative of its left-hand side is at least g , bounded uniformly away from zero, so the stated error is preserved on inversion. Finally, the functions of α appearing in the formula have bounded derivatives on a slightly smaller compact set; changing α by $O(1/N)$ changes the expression by $O(L/N)$, which is absorbed by its error term. $\square$

At a balanced composition, $A = \sqrt{k}$, $g = k-1$, and $D(\alpha) = D_k$, recovering the preceding center formula through its constant term. For $k=2$, $g = 2\sqrt{\alpha_1\alpha_2}$, recovering the binary bulk scale. Cauchy–Schwarz gives $g \leq k-1$, with equality only at the balanced composition, so the leading threshold coefficient s/g is minimized exactly at the center of each face.

3.2 Balanced bins and global modes

Theorem 3.6 (balancing within a support). *For $1/d \leq p < 1$, among bins with a prescribed set of positive coordinates and total N , probability is maximized precisely by the balanced count vectors. More precisely, transferring one unit from a coordinate b to another coordinate a , where $b \geq a+2$ and $a \geq 1$, strictly increases the bin probability.*

Proof. At $p = 1/d$ this follows from the ordinary multinomial formula. For $1/d < p < 1$, put $\sigma = p - r > 0$ and $v = r/\sigma > 0$. The positive refresh sum has the integral form

$$P_N(n) = \frac{\sigma^{N-1}}{dv} \int_0^\infty e^{-t} \prod_i B_{n_i}(vt) dt, \quad B_n(y) = \sum_{m=1}^n \binom{n-1}{m-1} \frac{y^m}{m!}. \quad (10)$$

For each $y > 0$, the sequence $B_n(y)$, $n \geq 1$, is strictly log-concave. We give an elementary proof of this instance of preservation of log-concavity by the binomial transform. Write

$$\frac{B_n(y)}{y} = b_{n-1}, \quad b_j = \sum_{i=0}^j \binom{j}{i} a_i, \quad a_i = \frac{y^i}{(i+1)!}, \quad c_j = \sum_{i=0}^j \binom{j}{i} a_{i+1}.$$

Pascal's identity gives $b_{j+1} = b_j + c_j$. The ratio c_j/b_j is the expectation of the strictly decreasing sequence $a_{i+1}/a_i = y/(i+2)$ under weights proportional to $\binom{j}{i}a_i$. When j increases to $j+1$, the likelihood ratio of the new and old unnormalized weights on the old support is $(j+1)/(j+1-i)$, increasing in i ; additionally a strictly positive weight appears at $i = j+1$. Thus these weights shift strictly toward larger indices. The expectation of a strictly decreasing sequence decreases strictly. This last implication follows directly by cross-multiplying the expectations and pairing terms with indices $i < \ell$, so no limiting probability argument is involved. Hence $b_{j+1}/b_j = 1 + c_j/b_j$ decreases strictly with j , proving the asserted strict log-concavity.

It follows that $B_{a+1}(y)B_{b-1}(y) > B_a(y)B_b(y)$ whenever $b \geq a+2$. Apply this inequality pointwise for $y = vt > 0$ in (10) and integrate. Repeated transfers give the conclusion. $\square$

Corollary 3.7 (global modes in the critical window). *Fix d . In the window of Theorem 3.2, if $t < a_d$, then for all sufficiently large N the only global maxima are the d vertices. If $t > a_d$, then the only global maxima are the fully supported balanced bins. These assertions are uniform on compact sets of t separated from a_d . At the exact full-center/vertex crossing, for all sufficiently large N the global maxima are exactly the vertices and the fully supported balanced bins, with both classes present.*

Proof. Theorem 3.6 reduces the possible modes to balanced face bins of support sizes $1, \dots, d$. For $2 \leq k \leq d$ their limiting log ratios to a vertex are $(k-1)(t-a_k)$. If $t < a_d$, these are all negative. If $t > a_d$, the full-center ratio exceeds the vertex ratio, and for $2 \leq k < d$ its limiting log ratio exceeds the k -face one by

$$(d-k)(t-a_d) + (k-1)(a_k-a_d) > 0.$$

Uniformity in Theorem 3.2 gives both assertions. At the exact full-center crossing, its window parameter tends to a_d by Theorem 3.1; every proper face-center ratio then has a limit strictly less than one. The same reduction proves the final claim. $\square$

Example 3.8 (an intermediate face can be a finite-size mode). The preceding asymptotic restriction cannot be replaced by a claim for every N . Take $d = 3$, $N = 4$, and $p = 10/17$, so $u = 7/20$. For an edge center and a full center respectively,

$$\frac{P_4(2, 2, 0)}{V_4} = 2u + 2u^2 + 2u^3 = \frac{4123}{4000} > 1, \quad \frac{P_4(2, 1, 1)}{V_4} = 6u^2 + 6u^3 = \frac{3969}{4000} < 1.$$

Theorem 3.6 shows that the three edge centers are the only global modes. Thus the common asymptotic window coexists with additional finite-size modal behavior.

4 A uniform simplex estimate and a boundary crossover

The compact-interior assumption in Theorem 3.5 excludes bins approaching a lower-dimensional face. We now remove that assumption from the discrete-to-Bessel estimate and resolve a transition in which several positive coordinates become small. The asymptotic transition proved below has at least two macroscopic coordinates; the one-macroscopic-coordinate case has a different persistence scale.

Put

$$\Phi(x) = \sum_{j \geq 0} \frac{x^j}{j!(j+1)!} = \frac{I_1(2\sqrt{x})}{\sqrt{x}}, \quad \Phi(0) = 1.$$

As before, $S_n(u)$ is the probability of the positive count vector n divided by the probability of one vertex, and $u = r/p$. All the statements are independent of the ambient number of colors once the support is fixed.

Lemma 4.1 (a uniform Laguerre–Bessel inequality). *For every integer $a \geq 1$ and real $w > 0$,*

$$0 \leq 1 - \frac{B_a(w)}{w\Phi(aw)} \leq \frac{w}{2}, \quad B_a(w) = \sum_{m=1}^a \binom{a-1}{m-1} \frac{w^m}{m!}. \quad (11)$$

Proof. The coefficient of $(aw)^j/[j!(j+1)!]$ in $B_a(w)/w$ is $\prod_{h=1}^j (1-h/a)_+$. Its deficit from one lies between zero and $j(j+1)/(2a)$, including for $j \geq a$. Consequently

$$0 \leq \Phi(aw) - \frac{B_a(w)}{w} \leq \frac{1}{2a} \sum_{j \geq 0} \frac{j(j+1)(aw)^j}{j!(j+1)!} = \frac{w}{2} \Phi(aw).$$

□

Theorem 4.2 (a uniform Bessel majorant on every simplex stratum). *Let $n_1, \dots, n_k$ be arbitrary positive integers with sum N , and let $0 < u < 1$, $v = u/(1-u)$. Define*

$$\mathcal{Q}_n(u) = (1-u)^{N-1} v^{k-1} \int_0^\infty e^{-t} t^k \prod_{i=1}^k \Phi(n_i v t) dt.$$

Then, with no lower bound on any n_i/N ,

$$0 \leq 1 - \frac{S_n(u)}{\mathcal{Q}_n(u)} \leq k(k+1)v + \frac{k}{2} v^2 \left(\sum_i \sqrt{n_i} \right)^2 \leq k(k+1)v + \frac{k^2}{2} N v^2. \quad (12)$$

Thus this approximation has relative error $O_k((\log N)^2/N)$ whenever $u = O(\log N/N)$, uniformly over every positive composition of N .

Proof. The positive refresh formula gives

$$S_n(u) = \frac{(1-u)^{N-1}}{v} \int_0^\infty e^{-t} \prod_i B_{n_i}(vt) dt.$$

Applying Lemma 4.1 to each factor, and then the product-deficit inequality, bounds the relative deficit by $(kv/2)\mathbb{E}[T]$, where T has density proportional to $e^{-t} t^k \prod_i \Phi(n_i v t)$. For the probability weights proportional to $x^j/[j!(j+1)!]$, the identity $\mathbb{E}[J(J+1)] = x$ implies

$$x \frac{\Phi'(x)}{\Phi(x)} = \mathbb{E}[J] \leq \sqrt{x}.$$

Integration by parts therefore gives

$$\mathbb{E}[T] = k+1 + \sum_i \mathbb{E} \left[\frac{n_i v T \Phi'(n_i v T)}{\Phi(n_i v T)} \right] \leq k+1 + \sqrt{v} \left(\sum_i \sqrt{n_i} \right) \sqrt{\mathbb{E}[T]}.$$

The boundary terms vanish: near zero the density is $O(t^k)$, while $\Phi(x) \leq e^{2\sqrt{x}}$ bounds the tail by a polynomial times $\exp(-t + 2\sqrt{vt} \sum_i \sqrt{n_i})$. This also proves finiteness of the integral and of every moment used above. Applying $2ab \leq a^2 + b^2$ to the final term yields $\mathbb{E}[T] \leq 2(k+1) + v(\sum_i \sqrt{n_i})^2$. The claimed estimates follow, the last one by Cauchy–Schwarz. □

4.1 Coordinates of order $N/(\log N)^2$

Fix integers $r \geq 2$ and $\ell \geq 1$, and put $k = r + \ell$. Consider r large counts $n_1, \dots, n_r$ and ℓ further positive counts $a_1, \dots, a_\ell$. Write

$$N = \sum_i n_i + \sum_j a_j, \quad M = \sum_i n_i, \quad L = \log N, \quad \beta_i = \frac{n_i}{M}, \quad \rho_j = \frac{a_j L^2}{N}.$$

The β_i and ρ_j below always refer to these actual counts. Define

$$A = \sum_{i=1}^r \sqrt{\beta_i}, \quad g = A^2 - 1, \quad K = r + \ell - 1, \quad H = \frac{r-1}{2} + 2\ell, \quad c = \frac{K}{g},$$

and

$$D_r(\beta) = \frac{A^{r/2+1} (\prod_i \beta_i)^{-3/4}}{(4\pi)^{(r-1)/2}}.$$

Theorem 4.3 (a crossover between neighboring simplex strata). *Fix $\varepsilon > 0$ and $R < \infty$. Assume $\beta_i \geq \varepsilon$ and $0 < \rho_j \leq R$. Uniformly for such count vectors and $c_0 L \leq z \leq C_0 L$, where $0 < c_0 < C_0 < \infty$ are fixed,*

$$S_{(n,a)}(z/N) = D_r(\beta) A^{2\ell} \frac{z^H e^{gz}}{N^K} \prod_{j=1}^{\ell} \Phi\left(\frac{A^2 a_j z^2}{N}\right) \left(1 + O\left(L^{-1} + \frac{L^2}{N}\right)\right). \quad (13)$$

The unique positive solution $u_{n,a}$ of $S_{(n,a)}(u) = 1$ consequently satisfies

$$\begin{aligned} Nu_{n,a} = cL - \frac{H}{g} \log L - \frac{1}{g} \left\{ \log(D_r(\beta) A^{2\ell}) + H \log c + \sum_{j=1}^{\ell} \log \Phi(A^2 c^2 \rho_j) \right\} \\ + O\left(\frac{\log L}{L} + \frac{L^2}{N}\right). \end{aligned} \quad (14)$$

The constants in the errors depend only on the displayed fixed parameters.

Proof. Let $v = u/(1-u)$ and $Y = Mv$. By Theorem 4.2, with relative error $O_k(v + Nv^2)$, the exact ratio equals

$$\frac{(1-u)^{N-1} v^{\ell-1}}{\prod_{i=1}^r n_i} \int_0^\infty e^{-t} t^\ell \prod_{i=1}^r \left[\sqrt{\beta_i Y t} I_1(2\sqrt{\beta_i Y t}) \right] \prod_{j=1}^{\ell} \Phi(a_j v t) dt. \quad (15)$$

In the stated regime $Y \asymp L$ and all $a_j v Y$ are bounded. Set $w = \sqrt{t}$. The exponential from the r large-coordinate factors is $\exp(A^2 Y - (w - A\sqrt{Y})^2)$. On $|w - A\sqrt{Y}| \leq \sqrt{Y}/2$, the large-coordinate Bessel arguments are bounded below by a positive constant times Y . Their usual large-argument expansions are uniform, with relative error $O(1/Y)$. The remaining smooth factor, apart from constants and powers of Y , is

$$w^{r/2+1+2\ell} \prod_j \Phi(a_j v w^2).$$

Divide this factor by its value at $w = A\sqrt{Y}$. On the same interval its second derivative is $O(1/Y)$, uniformly: $w/\sqrt{Y}$ lies in a fixed compact subset of $(0, \infty)$, the quantities $a_j v Y$ lie in a fixed

compact interval, and $\Phi \geq 1$ there. Taylor's theorem, followed by integration against the centered Gaussian, eliminates the linear term and gives relative error $O(1/Y)$.

For completeness, the part outside this interval is exponentially negligible uniformly. The bounds $I_1(x) \leq e^x$ and $\Phi(x) \leq e^{2\sqrt{x}}$ give a polynomial times

$$\exp\left(A^2Y - (w - A\sqrt{Y})^2 + 2w \sum_j \sqrt{a_j v}\right).$$

Here $\sum_j \sqrt{a_j v} = O(Y^{-1/2})$, so completing the square leaves a Gaussian centered only $O(Y^{-1/2})$ away from $A\sqrt{Y}$. On the excluded region its integral is exponentially small relative to the main term, even after the polynomial factors are included. The same argument also justifies extending the central Gaussian integrals to the full real line.

Evaluation of the leading Gaussian integral in (15) now gives

$$\begin{aligned} S_{(n,a)}(u) &= D_r(\beta) A^{2\ell} \frac{Y^H}{MK} e^{A^2Y + (N-1)\log(1-u)} \prod_j \Phi(A^2 a_j M v^2) \\ &\quad \times (1 + O(Y^{-1} + v + N v^2)). \end{aligned}$$

Since $\sum_j a_j/N = O(L^{-2})$, at $u = z/N$ we have

$$Y = z + O(L^{-1} + L^2/N), \quad A^2Y + (N-1)\log(1-u) = gz + O(L^{-1} + L^2/N).$$

Replacing Y by z , M by N , and $A^2 a_j M v^2$ by $A^2 a_j z^2/N$ in the other factors changes the expression by a relative $O(L^{-2} + L/N)$, because the arguments of Φ stay bounded and $\Phi \geq 1$. This proves (13).

The proper-run polynomial has nonnegative coefficients, zero constant coefficient, and positive coefficient $k!$ at u^{k-1} ; hence the crossing is unique. On the compact set of permitted β , both g and $D_r(\beta)$ are bounded above and bounded away from zero. Testing (13) at fixed sufficiently small and large multiples of L brackets the root uniformly. Its logarithm gives

$$gz + H \log z = KL - \log(D_r(\beta) A^{2\ell}) - \sum_j \log \Phi(A^2 \rho_j (z/L)^2) + O(L^{-1} + L^2/N).$$

First $z = cL + O(\log L)$, so the sum of logarithms on the right differs from its value at $z = cL$ by $O(\log L/L)$. Substituting this estimate into $\log z$ proves (14); inversion preserves the error because the derivative of $gz + H \log z$ is at least $g > 0$. $\square$

Corollary 4.4 (invisibility of subcritical rare counts). *Under the hypotheses of Theorem 4.3, if all $a_j = o(N/(\log N)^2)$, the sum involving Φ in (14) is $o(1)$. Thus, at fixed actual large composition β , the rare counts do not affect the expansion through its constant term. Their number ℓ still changes both its leading and its logarithmic coefficients.*

The scale $N/(\log N)^2$ here concerns approaching a simplex face with at least two macroscopic coordinates. It is not a replacement for the binary edge scale studied in the unpublished companion [2]: there is only one macroscopic coordinate at a binary endpoint, and then $g = 0$. The one-macroscopic-coordinate case is treated directly below and in Section 5, without using results from the companion. Nor does Theorem 4.3 assert an error bound when any ρ_j is unbounded.

Lemma 4.5 (exact insertion of a singleton color). *Let n have total size M and let $\mathcal{D} = u d/du$. Then*

$$S_{(n,1)}(u) = [2u + (M-1)u^2] S_n(u) + (u - u^2) \mathcal{D} S_n(u).$$

Proof. Insert the new color into a word of length M with j changes. The two end positions contribute a factor $2u$; the j change gaps contribute ju , and the $M - 1 - j$ repeat gaps contribute $(M - 1 - j)u^2$. Removing the unique new letter reverses this construction. Summing the resulting identity over all words proves the formula. $\square$

The singleton identity gives an exact finite-size check on the factor $A^2 z^2 / N$ contributed by one very small coordinate in (13). The finite-size inequalities above are proved algebraically; the asymptotic statements use real analysis and are not included in the accompanying Lean formalization.

4.2 Fixed rare counts near a vertex

The singular case of one large coordinate admits a different, finite polynomial limit. In this result the rare counts are fixed, and the large coordinate b , rather than the total count, is the asymptotic parameter.

Theorem 4.6 (a multicolor Laguerre limit at a vertex). *Fix $\ell \geq 1$ and positive integers $a_1, \dots, a_\ell$. Define*

$$H(y) = \prod_{i=1}^{\ell} B_{a_i}(y), \quad Q(y) = \sum_{i < j} B'_{a_i}(y) B'_{a_j}(y) \prod_{h \neq i, j} B_{a_h}(y).$$

The sum defining Q is zero when $\ell = 1$. Uniformly on compact subsets of $y > 0$, as $b \rightarrow \infty$,

$$S_{(b,a)}(\sqrt{y/b}) = H(y) + \frac{2\sqrt{y}}{\sqrt{b}}(H'(y) + yQ(y)) + O_a(b^{-1}). \quad (16)$$

Let $y_a > 0$ be the unique root of $H(y_a) = 1$, and put $c_a = \sqrt{y_a}$. The unique endpoint-crossing odds satisfy

$$u_{b,a} = \frac{c_a}{\sqrt{b}} - \frac{1 + y_a Q(y_a) / H'(y_a)}{b} + O_a(b^{-3/2}). \quad (17)$$

For one rare color this recovers the binary fixed-edge correction $-1/b$. For at least two rare colors the coefficient of $1/b$ is strictly less than -1 .

Proof. Fix rare-run counts $m_i \in \{1, \dots, a_i\}$, put $m = \sum_i m_i$, and let q be the number of runs of the large color. The rare run lengths can be chosen in $\prod_i \binom{a_i-1}{m_i-1}$ ways. Since successive large-color runs must be separated, $q \leq m + 1$; the large run lengths have $\binom{b-1}{q-1}$ choices. At $u = \sqrt{y/b}$, configurations with these run counts contribute $O_a(b^{(q-m-1)/2})$. There are finitely many possible m_i, q , independently of b , so only $q = m + 1$ and $q = m$ affect the expansion through order $b^{-1/2}$.

If $q = m + 1$, the run word starts and ends in the large color and alternates between large and rare colors. There are $m! / \prod_i m_i!$ rare orders. Using $\binom{b-1}{m} = b^m / m! + O_a(b^{m-1})$ and summing gives $H(y)$, with an $O_a(b^{-1})$ error.

If $q = m$, there are two possibilities. One endpoint is rare and the run word otherwise alternates, giving $2m! / \prod_i m_i!$ orders. These terms sum to $2\sqrt{y} H'(y) / \sqrt{b}$. Otherwise both endpoints are large and exactly one pair of consecutive rare runs is present. The members of that pair must have different colors. For an ordered pair of colors i, j , contracting the marked adjacent pair shows that there are $(m-1)! / [(m_i-1)!(m_j-1)! \prod_{h \neq i, j} m_h!]$ orders. Summing over the two orientations and all $i < j$ gives $2y^{3/2} Q(y) / \sqrt{b}$. Terms with $q \leq m - 1$ and the next terms of the binomial expansions are $O_a(b^{-1})$, uniformly for y in a positive compact set. This proves (16).

The polynomial H has positive coefficients, tends to zero at zero, and grows to infinity, so y_a exists uniquely and $H'(y_a) > 0$. The same positivity argument gives the unique crossing for each b . First bracket it using (16) to obtain $bu_{b,a}^2 \rightarrow y_a$. The finite run expansion may also be differentiated uniformly on this compact neighborhood; its derivative tends to $H'(y_a) > 0$. Substitution of $u = c_a/\sqrt{b} + \alpha/b$ into (16) then forces $\alpha = -1 - y_a Q(y_a)/H'(y_a)$, and the uniform remainder gives (17). If $\ell \geq 2$, all terms in $Q(y)$ are positive for $y > 0$, proving the strict inequality. $\square$

If all ℓ rare colors occur once, then $H(y) = y^\ell$ and $y_a = 1$, giving the explicit specialization

$$u_{b,1,\dots,1} = b^{-1/2} - \frac{\ell+1}{2b} + O_\ell(b^{-3/2}).$$

An independent exact formula in this case is

$$S_{(b,1,\dots,1)}(u) = \ell! \sum_{q=1}^{\min(b,\ell+1)} \binom{\ell+1}{q} \binom{b-1}{q-1} u^{q+\ell-1}.$$

Indeed, order the ℓ distinct rare letters, choose the q gaps among their $\ell+1$ end and interior gaps that contain large-color runs, and choose the positive lengths of those runs. In particular two singleton rare colors give

$$u_{b,1,1} = b^{-1/2} - \frac{3}{2}b^{-1} + O(b^{-3/2}).$$

This agrees with the exact polynomial $S_{(b,1,1)}(u) = 6u^2 + 6(b-1)u^3 + (b-1)(b-2)u^4$. Theorem 4.6 does not assert uniformity when the rare counts grow with b . Section 5 resolves the leading crossing uniformly for total rare count $A = o(b)$; uniform higher-order corrections in that full regime remain open here.

5 Uniform completion of the near-vertex regime

The fixed-count hypothesis in Theorem 4.6 can be removed at the level of the leading crossing threshold. The result is uniform over every rare composition with total mass $o(b)$. An accompanying example explains why the corresponding probability approximation need not be uniform: its error can diverge even while the relative error of the crossing threshold tends to zero.

Throughout this section, $\ell \geq 1$ is fixed, $a_1, \dots, a_\ell$ are positive integers, $A = \sum_i a_i$, and

$$H(y) = \prod_{i=1}^{\ell} B_{a_i}(y), \quad \kappa(y) = \frac{yH'(y)}{H(y)}.$$

Write y_a for the unique positive solution of $H(y_a) = 1$.

Lemma 5.1 (elasticity and concavity of the rare-count polynomial). *The polynomial H is convex, $\log H$ is concave on $(0, \infty)$, and κ is nondecreasing. Moreover,*

$$\frac{1}{A+1} \leq y_a \leq 1, \quad \kappa(y) \leq \ell + \sqrt{\ell A y}, \quad \kappa(y) \geq \max\{\ell, \tfrac{1}{2}\sqrt{A y}\} \quad (0 < y \leq 1). \quad (18)$$

Proof. Convexity follows directly from the nonnegative coefficients. For one factor, give $J \in \{0, \dots, a-1\}$ probabilities proportional to

$$r_j = \binom{a-1}{j} \frac{y^j}{(j+1)!}.$$

The successive ratio $R(j) = (j+1)r_{j+1}/r_j = y(a-1-j)/(j+2)$, with $R(a-1) = 0$, is nonincreasing. Summing the ratio identity gives $\mathbb{E}J = \mathbb{E}R(J)$ and $\mathbb{E}J^2 = \mathbb{E}[(J+1)R(J)]$. Therefore

$$\text{Var } J = \mathbb{E}J + \text{Cov}(J, R(J)) \leq \mathbb{E}J.$$

The covariance is nonpositive because, for an independent copy J' , twice the covariance equals $\mathbb{E}[(J - J')(R(J) - R(J'))] \leq 0$. Putting $M = J + 1$ and $\kappa_a = yB'_a/B_a = \mathbb{E}M$, we obtain

$$(\log B_a)'' = \frac{\text{Var } M - \mathbb{E}M}{y^2} \leq 0, \quad y\kappa'_a = \text{Var } M \geq 0.$$

This proves both qualitative assertions without invoking the locations of the zeros of a Laguerre polynomial.

The elementary differential equation $yB''_a + yB'_a = aB_a$ gives

$$ay = \kappa_a^2 + \text{Var } M + (y-1)\kappa_a.$$

Using $0 \leq \text{Var } M \leq \kappa_a - 1$ yields $\kappa_a \leq 1 + \sqrt{ay}$ and $\kappa_a^2 + y\kappa_a \geq ay + 1$. For $0 < y \leq 1$, these imply $\kappa_a \geq \max\{1, \sqrt{ay} - 1/2\} \geq \frac{1}{2}\sqrt{ay}$. Summing over the factors and applying Cauchy–Schwarz proves the elasticity bounds. Finally, $B_a(1) \geq 1$ and

$$B_a(y) \leq y\Phi(ay) \leq ye^{ay/2},$$

where $(j+1)! \geq 2^j$ proves the last inequality term by term. Thus $H(1) \geq 1$, whereas $H(1/(A+1)) \leq (A+1)^{-\ell} e^{A/(2(A+1))} < 1$. $\square$

Lemma 5.2 (an exact compound-binomial representation). *For $0 < u < 1$, put $v = u/(1-u)$. Let G have the gamma distribution with shape 2 and rate 1, let $E_1, \dots, E_{b-1}$ be independent rate-one exponential variables, and let $Z_1, \dots, Z_{b-1}$ be independent Bernoulli variables with parameter u . Take all these variables independent and set*

$$T = G + \sum_{j=1}^{b-1} Z_j E_j.$$

Then

$$S_{(b,a)}(u) = (1-u)^A \mathbb{E}[H(vT)], \quad \mathbb{E}e^{sT} = (1-s)^{-2} \left(1 + \frac{us}{1-s}\right)^{b-1} \quad (s < 1). \quad (19)$$

In particular, set $y = bu^2$, $\kappa = \kappa(y)$ and $s = \kappa/[bu(1-u)]$. Whenever $s < 1$,

$$(1-u)^A H(y) \leq S_{(b,a)}(u), \quad (20)$$

$$S_{(b,a)}(u) \leq (1-u)^A H(y) e^{-\kappa} (1-s)^{-2} \left(1 + \frac{us}{1-s}\right)^{b-1}. \quad (21)$$

Proof. The exact positive-refresh integral is

$$S_{(b,a)}(u) = \frac{(1-u)^{b+A-1}}{v} \int_0^\infty e^{-t} B_b(vt) H(vt) dt.$$

The integral of $e^{-t} B_b(vt)$ is $v(1+v)^{b-1}$. After normalization its density is a mixture of gamma densities with shape $J+2$, where $J \sim \text{Bin}(b-1, u)$. This is exactly the distribution of T , proving

the identity and its moment generating function. Since $\mathbb{E}T = 2 + (b-1)u$, $v\mathbb{E}T \geq bu^2 = y$. Convexity and monotonicity of H prove (20). Concavity of $\log H$ gives, for $w > 0$,

$$H(w) \leq H(y) \exp\left(\frac{\kappa}{y}(w-y)\right).$$

Insert $w = vT$ and use the moment generating function to prove (21). $\square$

Theorem 5.3 (all growing rare counts near a vertex). *For fixed ℓ , uniformly over all positive rare counts with $A = o(b)$, the unique positive crossing odds satisfy*

$$u_{b,a} = \sqrt{\frac{y_a}{b}} \left(1 + O_\ell\left(\sqrt{\frac{A+1}{b}}\right)\right), \quad \prod_i B_{a_i}(y_a) = 1. \quad (22)$$

A quantitative, unoptimized version is as follows. Put

$$K = \ell + \sqrt{\ell} + 1, \quad \varepsilon = \sqrt{\frac{A+1}{b}}, \quad \delta = 16K\varepsilon.$$

If $\delta \leq 1/2$, then

$$(1-\delta)\sqrt{y_a/b} \leq u_{b,a} \leq (1+\delta)\sqrt{y_a/b}. \quad (23)$$

No restriction such as $A(\log A)^2 = o(b)$ is needed.

Proof. The proper-run polynomial has positive coefficients, zero constant coefficient, and is unbounded at infinity; its crossing is unique. Put $u_0 = \sqrt{y_a/b}$ and $\kappa_0 = \kappa(y_a)$. At $u_+ = (1+\delta)u_0$ we have $u_+ < 1/4$. Monotonicity of κ and the lower bound in Lemma 5.1 give

$$\log H(bu_+^2) \geq 2\kappa_0 \log(1+\delta), \quad \frac{Au_0}{\kappa_0} \leq 2\sqrt{A/b} \leq 2\varepsilon.$$

Consequently, by (20),

$$\log S(u_+) \geq 2\kappa_0 \log(1+\delta) - \frac{Au_+}{1-u_+} \geq \kappa_0 \left(\frac{4}{3}\delta - 4\varepsilon\right) > 0.$$

At $u_- = (1-\delta)u_0$, write $y = bu_-^2$ and $\kappa = \kappa(y)$. The inequalities $y_a \geq 1/(A+1)$, $\delta \leq 1/2$, and (18) imply

$$u_- \leq \varepsilon, \quad bu_- \geq \frac{1}{2\varepsilon}, \quad s = \frac{\kappa}{bu_-(1-u_-)} \leq 2(2\ell + \sqrt{\ell})\varepsilon \leq 4K\varepsilon \leq \frac{1}{8}.$$

Using $\log(1+x) \leq x$, $-\log(1-s) \leq s/(1-s)$ and $u_-, s \leq 1/4$ in (21), and dropping the nonpositive term $A \log(1-u_-)$, gives

$$\begin{aligned} \log \frac{S(u_-)}{H(y)} &\leq -\kappa + \frac{2s}{1-s} + \frac{bu_-s}{1-s} \\ &= \kappa \left(\frac{1}{(1-u_-)(1-s)} - 1 \right) + \frac{2s}{1-s} \\ &\leq 5\kappa(u_- + s) \leq 25K\kappa\varepsilon. \end{aligned}$$

Here $\kappa \geq \ell \geq 1$ was used in the last two inequalities. Again by monotonicity of κ ,

$$\log H(y) \leq 2\kappa \log(1-\delta) \leq -2\kappa\delta.$$

Thus $\log S(u_-) \leq \kappa(-32K + 25K)\varepsilon < 0$. The two strict inequalities and uniqueness bracket the crossing. $\square$

Corollary 5.4 (an explicit shape law for growing rare counts). *Suppose $A \rightarrow \infty$, $A = o(b)$, and the actual proportions $\alpha_i = a_i/A$ stay in a fixed compact subset of the positive ℓ -simplex. Put $h = \sum_i \sqrt{\alpha_i}$. Then*

$$u_{b,a} \sim \frac{\ell \log A}{2h\sqrt{Ab}}.$$

More precisely, the polynomial root satisfies

$$2h\sqrt{Ay_a} = \ell \log A - \frac{\ell}{2} \log \log A + \frac{\ell}{2} \log \left(\frac{8\pi h}{\ell} \right) + \frac{3}{4} \sum_i \log \alpha_i + O_\ell \left(\frac{\log \log A}{\log A} + \frac{(\log A)^2}{A} \right).$$

Replacing $\sqrt{Ay_a}$ on the left by $\sqrt{Ab}u_{b,a}$ incurs an additional $O_\ell(\log A \sqrt{A/b})$ error.

Proof. Lemma 4.1, uniformly when $y \rightarrow 0$, gives $H(y) = y^\ell \prod_i \Phi(a_i y)(1 + O_\ell(y))$. Write $x = \sqrt{Ay}$. Bracketing first at fixed small and large multiples of $\log A$ shows that the root has $x \asymp \log A$. The standard large-argument expansion of I_1 then gives uniformly over the stated proportions

$$H(x^2/A) = \frac{x^{\ell/2} e^{2hx}}{A^\ell (4\pi)^{\ell/2} (\prod_i \alpha_i)^{3/4}} (1 + O_\ell(x^{-1} + x^2/A)).$$

Taking logarithms, substituting $x = (\ell/(2h)) \log A + O_\ell(\log \log A)$ in the logarithmic term, and using Theorem 5.3 proves the claims. The constants also depend on the fixed compact set of proportions. $\square$

Proposition 5.5 (probability approximation can fail while roots agree). *Let $a = \lfloor b/\log b \rfloor$, let y_a solve $B_a(y_a) = 1$, and set $u_0 = \sqrt{y_a/b}$. Then*

$$H(y_a) = B_a(y_a)^2 = 1, \quad S_{(b,a,a)}(u_0) \rightarrow \infty, \quad \frac{u_{b,a,a}}{u_0} \rightarrow 1.$$

Thus crossing roots admit a uniform approximation in a regime where the corresponding relative probability approximation fails: $S(u) \sim H(bu^2)$ is false as a uniform assertion over $A = o(b)$.

Proof. Retain only run words that start and end in the large color and have exactly one adjacent pair of distinct rare colors. The run enumeration in Theorem 4.6 shows that their contribution is the polynomial $2uyB'_a(y)^2$, with each term of total rare-run count m additionally multiplied by $\prod_{j=1}^{m-1} (1 - j/b)_+$. Give J probabilities proportional to $\binom{a-1}{j} y^j/j!$, the coefficients of $B'_a(y)$. Its differential equation $y(B'_a)'' + (1+y)(B'_a)' - (a-1)B'_a = 0$ implies $\mathbb{E}J^2 + y\mathbb{E}J = (a-1)y$; hence $\mathbb{E}J \leq \sqrt{ay}$ and $\mathbb{E}J^2 \leq ay$. For two independent copies, $m = J_1 + J_2 + 2$, so

$$\mathbb{E}[m(m-1)] \leq 4ay + 6\sqrt{ay} + 2.$$

The clipped-product deficit bound therefore proves

$$S_{(b,a,a)}(u_0) \geq \frac{2\kappa_a(y_a)^2}{bu_0} \left(1 - \frac{4ay_a + 6\sqrt{ay_a} + 2}{2b} \right).$$

The preceding corollary for one rare color gives $\sqrt{ay_a} \sim \frac{1}{2} \log a$, and the elasticity bounds give $\kappa_a(y_a) \sim \sqrt{ay_a}$. The last display is asymptotic to a lower bound $\sqrt{a/b} \log a \sim \sqrt{\log b}$, which diverges. The root assertion follows directly from Theorem 5.3, since $2a = o(b)$. $\square$

These results complete the leading near-vertex threshold for any fixed number of rare colors. They do not classify every finite-size global mode, nor do they assert a uniform higher-order root expansion for all growing rare counts. The analytic arguments in this section have written proofs and are not included in the finite Lean formalization.

6 Spatial superlevel sets and weak-field face phases

The interior-composition estimate also determines the geometry of the bins whose individual probabilities exceed the vertex probability. We give the result in a weak external field, since this perturbation exhibits additional face phases that the symmetric critical window suppresses.

Fix d and the symmetric transition model of Section 3. For $h = (h_1, \dots, h_d) \in \mathbb{R}^d$, define the *tilted occupation law*

$$\tilde{P}_{N,h}(n) = \frac{e^{h \cdot n/N} P_N(n)}{\sum_{m_1 + \dots + m_d = N} e^{h \cdot m/N} P_N(m)}. \quad (24)$$

Thus the field on each step is h_i/N . This is a Gibbs tilt of the occupation law; it is not a claim that the original homogeneous transition probabilities have acquired this bias. The normalization cancels in every bin comparison. A modal phase below specifies which face contains the largest individual bin probabilities; it does not assert concentration of the total probability mass on that face. Put $L = \log N$, retain

$$u_N(t) = \frac{L - \frac{1}{2} \log L + t}{N}, \quad p_N(t) = \frac{1}{1 + (d-1)u_N(t)},$$

and write $h_* = \max_i h_i$ and $\tilde{V}_{N,h} = \max_i \tilde{P}_{N,h}(N e_i)$. For a support $S \subseteq \{1, \dots, d\}$ of cardinality $k \geq 2$, set

$$\bar{h}_S = \frac{1}{k} \sum_{i \in S} h_i, \quad B_S(t, h) = (k-1)(t - a_k) + \bar{h}_S - h_*, \quad H_S = \{y \in \mathbb{R}^S : \sum_{i \in S} y_i = 0\}.$$

Euclidean measure on H_S is understood, and $\omega_s = \pi^{s/2}/\Gamma(1 + s/2)$ denotes the volume of its s -dimensional unit ball when $s = k-1$.

Theorem 6.1 (Gaussian superlevel geometry). *For a count vector with support exactly S , put*

$$y_i = \sqrt{L} \left(\frac{n_i}{N} - \frac{1}{k} \right), \quad i \in S.$$

Uniformly when t, h, y range over fixed compact sets, with $y \in H_S$,

$$\frac{\tilde{P}_{N,h}(n)}{\tilde{V}_{N,h}} = \exp \left(B_S(t, h) - \frac{k^2}{4} \sum_{i \in S} y_i^2 + o(1) \right). \quad (25)$$

Let $M_{N,S}(t, h)$ be the number of bins of support exactly S whose tilted probability is strictly greater than $\tilde{V}_{N,h}$. Then, uniformly for (t, h) in compact sets,

$$\frac{L^{(k-1)/2}}{N^{k-1}} M_{N,S}(t, h) \longrightarrow \frac{\omega_{k-1}}{\sqrt{k}} \left(\frac{4[B_S(t, h)]_+}{k^2} \right)^{(k-1)/2}, \quad [x]_+ = \max(x, 0). \quad (26)$$

If $B_S(t, h) < 0$, there are no such bins for all sufficiently large N . At $B_S(t, h) = 0$, the normalized count tends to zero; no eventual emptiness is asserted at that boundary.

Proof. With $\alpha_i = 1/k + y_i/\sqrt{L}$ and $\sum_i y_i = 0$, Taylor expansion on any fixed compact set in H_S gives

$$\left(\sum_i \sqrt{\alpha_i} \right)^2 = k - \frac{k^2}{4L} \sum_i y_i^2 + O(L^{-3/2}), \quad D(\alpha) = D_k(1 + o(1)).$$

Insert these expressions and $z = L - \frac{1}{2} \log L + t$ into Theorem 3.5. Since $\log D_k = -(k-1)a_k$, the logarithm of the untilted ratio is

$$(k-1)(t - a_k) - \frac{k^2}{4} \sum_i y_i^2 + o(1).$$

The logarithm of the field factor relative to the largest vertex is $h \cdot \alpha - h_* = \bar{h}_S - h_* + o(1)$, uniformly on the stated compact sets. This proves (25).

The local estimate must also exclude superlevel bins near the boundary of the face. We supply that step rather than assuming that a local limit controls all bins. Fix compact sets of t, h , and choose a sufficiently large constant R . Starting from any positive count vector on S , successively transfer one unit from a coordinate b to a coordinate a when $b \geq a + 2$. These transfers preserve the support and terminate at a balanced vector. By Theorem 3.6, the untilted probability strictly increases at every transfer. Each transfer changes the scaled vector y by norm $\sqrt{2L}/N$ and decreases its squared norm. If the starting vector has $\|y\| > R$, the first vector along this path with norm at most R has norm $R + o(1)$; it lies in a compact subset of the relative interior of the face. Its untilted log ratio to a vertex is at most

$$(k-1)(t - a_k) - k^2 R^2 / 4 + o(1).$$

This is an upper bound for the starting vector's untilted log ratio. For every composition α , $h \cdot \alpha - h_* \leq 0$. Consequently R can be chosen uniformly so that every tilted superlevel bin satisfies $\|y\| \leq R$. The same argument, with a larger R if necessary, bounds all face maximizers when comparison is made with its balanced bin.

Inside this ball the logarithms in (25) converge uniformly. For any $\delta > 0$, their positive set is sandwiched between the lattice points in balls of squared radii $4[B_S - \delta]_+/k^2$ and $4[B_S + \delta]_+/k^2$, with the inner ball interpreted as empty when $B_S \leq \delta$. The scaled count vectors form a translate of the lattice

$$\frac{\sqrt{L}}{N} \{z \in \mathbb{Z}^S : \sum_i z_i = 0\} \quad \text{in } H_S.$$

Its covolume is $\sqrt{k}(\sqrt{L}/N)^{k-1}$: the basis $e_i - e_k$, $1 \leq i < k$, has Gram determinant k . Lattice Riemann sums over balls therefore give the stated volume constant. Letting $\delta \downarrow 0$ proves (26), including $B_S = 0$. The radii stay bounded on compact parameter sets, so the same sandwich proves uniform convergence. If $B_S < 0$, the uniform profile on the confining ball is strictly below one for large N , proving emptiness. $\square$

At zero field, all supports of the same size have the same count, so the right-hand side of (26) is multiplied by $\binom{d}{k}$ when counting all bins with k positive coordinates. For $k = 2$, writing $y = (y_1, -y_1)$ gives the profile $\exp(t - a_2 - 2y_1^2)$ and the one-face count constant $\sqrt{2[t - a_2]_+}$, agreeing with the free binary window after its centering convention is taken into account.

Theorem 6.2 (weak-field modal phase diagram). *Define the face scores*

$$\lambda_S(t, h) = \begin{cases} h_i, & S = \{i\}, \\ (|S| - 1)(t - a_{|S|}) + \bar{h}_S, & |S| \geq 2. \end{cases}$$

The largest bin weight, expressed relative to the untilted vertex before normalizing (24), satisfies

$$\log \max_n \left\{ e^{h \cdot n / N} \frac{P_N(n)}{V_N} \right\} \longrightarrow \max_{\emptyset \neq S \subseteq \{1, \dots, d\}} \lambda_S(t, h). \quad (27)$$

For fixed (t, h) , every global mode has, for all sufficiently large N , a support maximizing the limiting score. If the maximizing support is unique, all global modes have exactly that support. On any selected support S with $|S| \geq 2$, their locations satisfy

$$\max_{i \in S} \left| \frac{n_i}{N} - \frac{1}{|S|} \right| = o((\log N)^{-1/2}).$$

These assertions do not identify the maximizing integer vectors exactly, and equality of limiting scores does not assert finite- N coexistence.

If the fields are sorted $h_1 \geq \dots \geq h_d$, the score maximization reduces to the upper envelope of the d lines

$$\lambda_1(t) = h_1, \quad \lambda_k(t) = (k-1)(t - a_k) + \frac{1}{k} \sum_{i=1}^k h_i, \quad 2 \leq k \leq d. \quad (28)$$

Away from ties, the winning support size is nondecreasing as t increases. If the fields are strictly ordered, the corresponding supports are the nested sets $\{1, \dots, k\}$. Every support size $k \in \{2, \dots, d-1\}$ can be the unique winning size on an open interval for suitable fixed fields. Thus proper-face phases can persist asymptotically under a field of order $1/N$ per step.

Proof. For each fixed support, the confinement argument in the preceding proof applies also to its maximum, because the field factor is bounded above and below uniformly in N . On that compact scaled region, the profile relative to the untilted vertex is $\exp(\lambda_S - k^2 \|y\|^2/4 + o(1))$. Balanced integer vectors have $y = o(1)$, so the face maximum converges to e^{λ_S} . There are finitely many supports. Taking their maximum proves (27), and the positive gap to any nonmaximizing score excludes its support for large N . If a sequence of maximizers within a selected face had scaled norm bounded away from zero, the uniform Gaussian profile would make it strictly smaller than that face's balanced bin. This proves the location statement.

Among supports of a given size k , the largest field average is the average of the k largest fields. This proves (28). For $t_1 < t_2$, a line of strictly larger slope cannot win at t_1 and lose to a line of smaller slope at t_2 : subtracting the two comparisons would give $(k_1 - k_2)(t_2 - t_1) \leq 0$. Hence the winning size is nondecreasing away from ties.

To realize a prescribed $2 \leq k < d$, set $h_1 = \dots = h_k = 0$ and $h_{k+1} = \dots = h_d = -H$. Choose $t > a_k$ sufficiently close to a_k so that $t < a_{k-1}$ if $k > 2$. The k -face score is positive; all smaller nonvertex face scores are negative, and the largest vertex score is zero. For $j > k$, the largest score is

$$(j-1)(t - a_j) - H \frac{j-k}{j}.$$

Taking H sufficiently large makes each of these strictly smaller than the k -face score. Every other support of size k includes a negative field and also loses. All comparisons are strict, so they persist on an open interval of t . Small perturbations can make the fields strictly ordered without removing the phase. $\square$

Corollary 6.3 (one field realizes the full face hierarchy). *Fix $d \geq 2$ and $H > 3/4$, and choose*

$$h_i = -H(i-1)^2, \quad 1 \leq i \leq d.$$

In particular $h_i = -(i-1)^2$ is a single explicit choice valid in every fixed dimension. Put

$$c_k = (k + \frac{1}{2}) \log k - \frac{k-1}{2} \log(4\pi), \quad \tau_k = c_k - c_{k+1} + \frac{H(4k-1)}{6} \quad (1 \leq k < d),$$

and set $\tau_0 = -\infty$, $\tau_d = +\infty$. Then $\tau_1 < \dots < \tau_{d-1}$, and for every $k \in \{1, \dots, d\}$ the support $\{1, \dots, k\}$ is the unique winning support throughout the nonempty interval $\tau_{k-1} < t < \tau_k$. Consequently, for each fixed t in this interval and all sufficiently large N , every global mode of the tilted law has exactly those k positive coordinates.

Proof. Here $c_1 = 0$ and $c_k = -(k-1)a_k$ for $k \geq 2$. Since $k^{-1} \sum_{i=1}^k (i-1)^2 = (k-1)(2k-1)/6$, the lines in (28) have intercepts $c_k - H(k-1)(2k-1)/6$. The intersection of lines k and $k+1$ is therefore $t = \tau_k$. Extend c_k to the displayed smooth function $c(x)$ for $x \geq 1$. Direct differentiation gives

$$0 < c''(x) = \frac{x-1/2}{x^2} \leq \frac{1}{2}.$$

Integrating this bound twice over unit intervals yields $c_{k+2} - 2c_{k+1} + c_k \leq 1/2$. Hence

$$\tau_{k+1} - \tau_k = \frac{2H}{3} - (c_{k+2} - 2c_{k+1} + c_k) > 0.$$

For $\tau_{k-1} < t < \tau_k$, summing consecutive differences $\lambda_{j+1}(t) - \lambda_j(t) = t - \tau_j$ shows that line k strictly exceeds every earlier and later line. The fields are strictly ordered, so its unique best support is $\{1, \dots, k\}$. Apply Theorem 6.2 to obtain the finite- N conclusion. $\square$

Example 6.4 (a persistent triangular edge phase). For $d = 3$ and $h = (0, 0, -H)$, the three competing maximal scores are

$$0, \quad t - a_2, \quad 2(t - a_3) - H/3.$$

If $H > 6(a_2 - a_3)$, the edge with support $\{1, 2\}$ wins strictly for

$$a_2 < t < H/3 + 2a_3 - a_2.$$

For every fixed t in this interval, all global modes of the tilted law lie on that edge for sufficiently large N . This differs from the untilted law, where proper-face modes disappear from the asymptotic critical window. It does not strengthen the finite-size counterexample of Example 3.8 into an untilted asymptotic claim.

7 Verification and remaining questions

The proofs above are analytic and combinatorial. Independent finite checks are supplied by the standard-library Python scripts accompanying this draft. The script `verify_geometry.py` compares 6,733 exact rational probabilities in 237 distributions with dynamic programming; 158 distributions also agree with exhaustive enumeration. It checks normalization, means, covariances, and endpoint parameters for $d = 2, 3, 4$. The script `verify_simplex_threshold.py` checks the positive refresh formula, balancing transfers, the finite-size edge-mode example, and crossing expansions. The scripts `verify_frontier_boundary.py` and `verify_frontier_modes.py` check the uniform comparison, rare-coordinate formulas, spatial profiles, and weak-field assertions against finite calculations. Finally, `verify_vertex_completion.py` contains 462 exact identity comparisons, 200 rational elasticity checks, 16 explicit root-bracket checks, and four large growing-rare-count stress cases. The associated records are stored under `data/`. Decimal asymptotic checks are numerical corroboration, not interval certificates or substitutes for the written proofs.

For fixed support size, the leading near-vertex crossing is now uniform throughout $A = o(b)$. A uniform higher-order expansion throughout that whole regime remains open here. Other

remaining questions include a growing number of colors, nonuniform switching, and the complete finite-size mode diagram outside the proved window. The explicit finite-size edge modes and the divergent probability approximation in Proposition 5.5 delimit the results: they prevent stronger interpretations of the corresponding asymptotic claims. Before submission, the proposed refinements require independent mathematical review and completion of the literature comparison.

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